- ***Support for a variety of hardware:*** Contiki can be run on a variety of low-power devices, which are easily available online.
- ***Large community support:*** One of the important advantages of using Contiki is the availability of an active community of developers. So when you have some technical issues to be solved, these community members make the problem solving process simple and effective.

The major features of Contiki are listed below.

- ***Memory allocation:*** Even the tiny systems with only a few kilobytes of memory can also use Contiki. Its memory efficiency is an important feature.
- ***Full IP networking:*** The Contiki OS offers a full IP network stack. This includes major standard protocols such as UDP, TCP, HTTP, 6lowpan, RPL, CoAP, etc.
- ***Power awareness:*** The ability to assess the power requirements and to use them in an optimal minimal manner is an important feature of Contiki.
- ***The Cooja network simulator*** makes the process of developing and debugging software easier.
- The availability of the Coffee Flash file system and the Contiki shell makes the file handling and command execution simpler and more effective.

TinyOS

TinyOS is an open source operating system designed for low-power wireless devices. It has a vibrant community of users spread across the world from both academia and industry. The popularity of TinyOS can be understood from the fact that it gets downloaded more than 35,000 times in a year.

TinyOS is very effectively used in various scenarios such as sensor networks, smart buildings, smart meters, etc. The main repository of TinyOS is available at *https://github.com/tinyos/tinyos-main.*

TinyOS is written in nesC which is a dialect of C. A sample code snippet is shown below:

```
configuration Led {
   provides {
     interface LedControl;
   }
    uses {
     interface Gpio;
   }
 }
 implementation {

   command void LedControl.turnOn() {
     call Gpio.set();
   }

   command void LedControl.turnOff() {
     call Gpio.clear();
   }

 }
```

Zephyr

Zephyr is a real-time OS that supports multiple architectures and is optimised for resource-constrained environments. Security is also given importance in the Zephyr design.

The prominent features of Zephyr are listed below:

- Support for 150+ boards.
- Complete flexibility and freedom of choice.
- Can handle small footprint IoT devices.
- Can develop products with built-in security features.

This article has introduced readers to a list of four OSs for the IoT, from which they can select the ideal one, based on individual requirements. END

References

[1] Ubuntu Core: *https://ubuntu.com/core*
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